Shoreline_All: Lightweight Shoreline Mapping by Leveraging Airborne Point Clouds to Address Data Sparsity or Missing in Vegetated Areas via Coastal Profile Method Fusion

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***Abstract*—Prevailing methods, whether based on imagery or point clouds, often fail to account for data sparsity or loss caused by vegetation occlusion. Moreover, while these methods are applicable to all coastal types, this generality precludes guarantees of depth accuracy and a high degree of data-algorithm compatibility. To address these limitations, this article introduces Shoreline_All, a novel framework. The proposed framework comprises four core modules: 1) vegetation detection and filling (VDF) module: introduces criteria for detecting vegetation point clouds and a novel algorithm for filling point cloud attributes within the residual spaces beneath vegetation; 2) classification, extraction, and processing (CEP) module: classifies coastal point clouds into land and water categories based on laser echo intensity, accurately extracts the land-ward and water-ward boundary of water and land point clouds, and subsequently fuses and offsets them; 3) precision transition (PT) module: infers the positional attributes of specific points from elevation characteristics within small sea-area segments, based on the principle that water flows from high to low, followed by further optimization; and 4) collaboration and smoothing (CAS) module: integrates and enhances the widely recognized coastal profile method (CPM), and a dedicated algorithm is then applied for smoothing and fitting the final shoreline point cloud. Extensive experiments were conducted on three distinct sea-area point clouds—encompassing various shoreline types. Compared with six state-of-the-art methods, our framework demonstrated superior accuracy, robustness, generalization capability, and lighter computational weight. Specifically, compared to the worst-performing baseline, our method improved shoreline accuracy by 0.4778, 0.4765, and 0.2773 m across the three datasets, while reducing processing time by 97, 140, and 58 s, respectively. Shoreline_All provides a new technical means for mapping shorelines, nautical charting, assessing coastal erosion, evaluating sea-level elevation variations, and supporting other critical marine management activities.**

***Index Terms*—Airborne LiDAR point clouds, coastal profile method (CPM), point cloud intensity value, shoreline extraction, vegetation occlusion.**

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I. Introduction

THE shoreline serves as a scientific abstraction and definitive boundary between ocean and land. Its precise definition, however, varies among countries and regions. Common definitions include the multiyear mean high-tide line, the instantaneous land–water interface, and the multiyear mean low tide line. Comprehensive surveys indicate that the multiyear mean high-tide line is the most widely adopted standard globally. Oceans cover approximately 71% of the Earth's surface, making the land–sea interface a critical zone for human habitation and activity. This extensive shoreline borders vast stretches of terrestrial environment, with vegetated coastal areas representing a significant portion of the total coastal zone. The spatiotemporal information of the shoreline reveals multidimensional physical and spatial dynamics, including changes in sea-level elevation, coastal erosion and evolution, variations in the extent of mudflats and wetlands, and the demarcation of charts and navigational boundaries. Among these applications, coastal erosion and evolution assessment is particularly vital. It reflects the susceptibility and fragility of shorelines and provides timely, reliable risk mitigation data for local communities, making it an important and active research topic [1], [2], [3], [4]. The effectiveness of all such applications fundamentally depends on acquiring accurate spatiotemporal shoreline data [5]. Consequently, employing scientific methodologies to obtain precise spatial characteristics of shorelines has become a priority in societal development.

Early shoreline mapping relied primarily on the interpretation of aerial imagery [6], [7], [8], which spurred the development of numerous related methods [9], [10], [11], [12], [13], [14], [15], [16]. These advances facilitated the progress of image-based shoreline extraction techniques and supported various shoreline-dependent applications during this period [17], [18], [19], [20], [21]. However, data acquisition from aerial or satellite imagery is susceptible to environmental, climatic, cloud cover, and illumination conditions. With the advancement of LiDAR technology, which enables precise acquisition of 3-D point coordinates, airborne LiDAR has gradually emerged as a new approach for shoreline information detection [22]. After decades of development, the use of airborne point clouds for shoreline mapping has generally evolved along three main directions.

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1) Direct Conversion Based on Tide Level Elevation: Methods under this category construct various processing and conversion techniques applied directly to the original sea-area point cloud, using the multiyear mean high-tide line elevation to output the shoreline point cloud directly.

2) Water–Land Boundary Transformation: These methods compute the boundary between water and land areas and subsequently convert it into a shoreline using a transformation model.

3) Coastal Classification-Based Extraction: These approaches classify coastal types and design tailored shoreline extraction methods according to the distinct characteristics of each category. The first type represents an early and relatively mature class of extraction algorithms [23], [24], [25], [26]. These typically process the entire sea-area point cloud, confronting challenges associated with large data volumes—coastal zones often contain millions of points. Executing a stepwise algorithm that requires searching for individual points across the entire dataset can lead to exponential computational costs, typically as a function of the point cloud size. Since such methods often involve multiple algorithmic stages, the aggregate computational burden becomes substantial. In addition, some researchers have constructed digital elevation models (DEMs) from original sea-area point clouds to extract shorelines. For instance, Yousef et al. [27] employed a point cloud-derived DEM combined with multistage morphological algorithms to determine shoreline positions relative to tidal datums. Similarly, Liu et al. [28] generated a DEM from point clouds and intersected it with tidal reference planes to produce a binary land–water classification, subsequently applying a series of image processing techniques—including region grouping, labeling, dual-scan analysis, and mathematical morphology—to accurately extract the shoreline. Generating a DEM from sea-area point clouds and subsequently identifying the mean high water (MHW) elevation contour represents a relatively novel approach. However, the DEM generation process itself is complex, computationally intensive, and prone to introducing errors. Moreover, such methods are typically designed for universal application across all coastal types, despite significant variations in terrain and feature distribution. These variations often manifest as abrupt elevation changes or noisy data points, which can exceed the compatibility of generalized algorithms. Consequently, extraction is liable to in accuracy or failure when local data characteristics deviate from algorithmic assumptions. Studies such as those cited in [29] and [30] have contributed to the development of shoreline extraction from airborne point clouds. Nevertheless, analysis reveals that these methods generally do not account for the problem of sparse or missing data due to vegetation occlusion—a significant oversight that can substantially compromise extraction accuracy in vegetated coastal zones. Furthermore, the high computational demand of these approaches, coupled with their generalized design for all coast types, increases the risk of data-method mismatch. As a result, the robustness of such methods requires further improvement.

The second category of methods involves extracting the land–water boundary and converting it into a shoreline point cloud using a transformation model. Various algorithms have been developed to derive this boundary [31], [32], [33], [34], [35]. For instance, Robertson W et al. [31] classified wet/dry lines (e.g., high water line, HWL) from topographic point cloud data and integrated them with tidal data—such as MHW and mean higher high water (MHHW)—to derive shorelines with errors below 5.4 m. The accuracy of the resulting digital terrain model (DTM) is crucial for coastal analysis, and this approach demonstrates good representativeness by utilizing sea-area point clouds to generate the DTM. Other techniques in this category involve converting point clouds into image maps and extracting features such as coastal slope, thereby advancing the extraction of instantaneous land–water boundaries and supporting applications that require such data. However, these methods generally overlook the presence of coastal vegetation. In areas with extensive vegetation, significant errors—or even failures—may occur in boundary delineation. Further analysis reveals that they primarily distinguish land from water based on extracted features but lack deeper mechanisms to ensure boundary accuracy in complex cases. In response to the substantial variations in coastal terrain and land cover—particularly obstructions caused by irregular geomorphic features in transitional zones—a third category of shoreline-specific classification and extraction methods has emerged. Researchers such as Ziheng Yang [36], Li et al. [22], H. M. A. van der Werff [37], and Bengoufa [38] have developed tailored approaches for specific shoreline types (e.g., vegetated, rocky, or sandy coasts), demonstrating markedly improved accuracy in their respective environments. Nevertheless, these methods still often fail to adequately address the issue of data sparsity or omission due to vegetative occlusion, and there remains room for improvement in both extraction accuracy and computational efficiency.

In summary, current airborne LiDAR-based shoreline extraction methods can be broadly categorized into these approaches: those are applicable to all types of shorelines. Those that first classify land and water boundaries before converting them into shorelines, and those that extract shorelines directly based on specific coastal typologies. These diverse methodologies have advanced point cloud-based shoreline extraction technology by addressing the challenge from multiple perspectives, significantly contributing to the field's development. Despite recent advancements, most existing methods fail to account for the obstruction caused by coastal vegetation, which can obscure the land–water interface and lead to data loss, significantly compromising shoreline extraction accuracy. Furthermore, in methods that classify land and water bodies or process different shoreline types separately, the land–water boundary extraction lacks high precision, and no optimization techniques are applied. As a result, these methods suffer from limited accuracy, robustness, and generalizability. This study addresses the challenge of missing data due to vegetation obstruction in coastal areas by proposing a novel algorithm for land–water boundary extraction. The method enhances boundary accuracy and further refines shoreline precision by integrating results from the current

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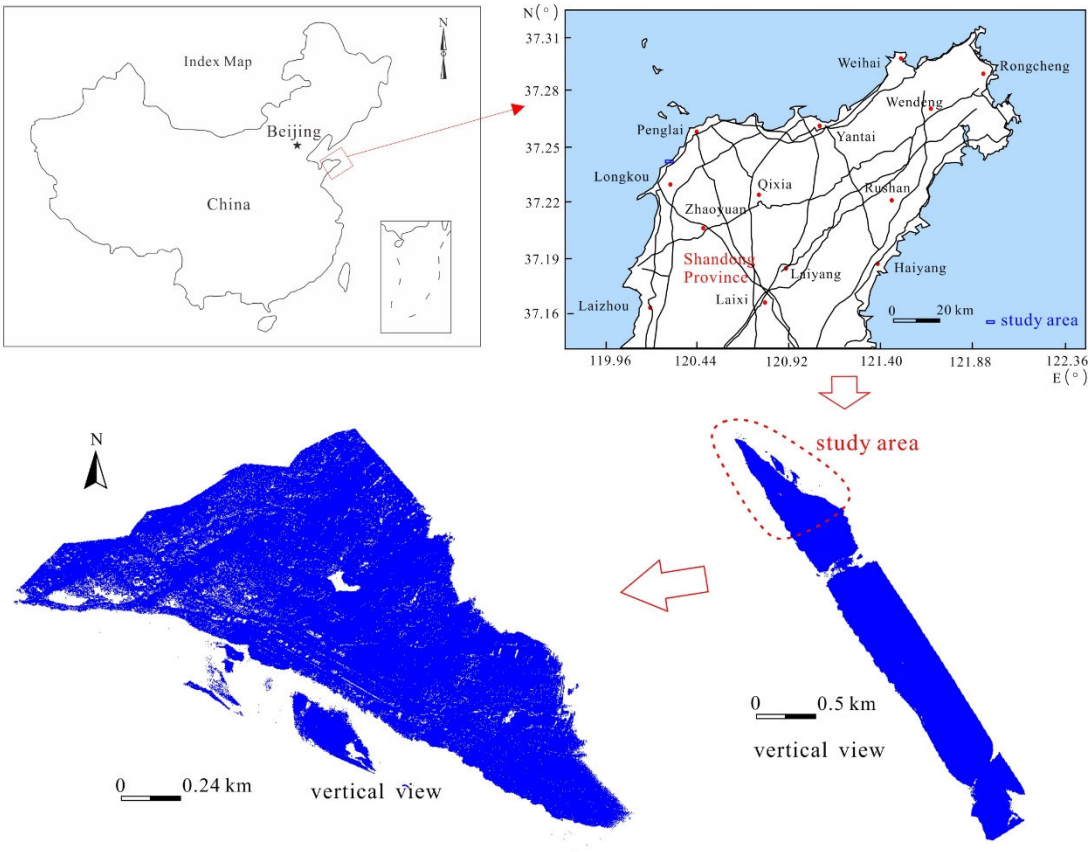


Fig. 1. Study area location and corresponding point cloud data.

best-performing coastal profile technique. In addition, a new criterion for fitting and interpolating shoreline point cloud data is introduced, ensuring smoother, more continuous, and natural shoreline representations. The proposed approach is applicable to diverse shoreline types and achieves higher extraction accuracy compared to existing methods. It offers valuable insights for shoreline extraction research and related studies related to shoreline. Furthermore, this method provides an advanced technological solution for shoreline mapping, management, and coastal conservation and development.

II. Data and Method

A. Overview of the Dataset and Method

This study focuses on a representative coastal region in Longkou City, Shandong Province. The main airborne LiDAR point cloud data were collected for this area, comprising 16 067 122 points (file size: 365 MB). The coordinate system is defined as follows, *x*-axis: horizontal right direction, *y*-axis: horizontal inward direction, *z*-axis: vertical upward direction. The dataset spans a spatial extent defined by the following coordinates in this system, the bottom-left (minimum) coordinate is x = 513 757.156, y = 3 372 091.500, z = 84.339, the top-right(maximum) coordinate is x = 522 890.437, y = 3372 078.250, z = 60.169. From the coastal point cloud data, a representative section of the shoreline was selected as the study area, encompassing diverse marine environments such as vegetated zones or sandy regions. The dataset for this study area comprises 772 243 points (file size: 21.58 MB). Within the given coordinate system, the bottom-left (minimum) coordinate is x = 513 757.156, y = 3 372 091.5, z = 84.339, and the top-right (maximum) coordinate is x = 51 709.938, y = 3 372 833.25, z = 111.05. Each point in the dataset contains 3-D spatial coordinates, laser echo intensity, laser echo frequency, and RGB color values, the geographical location of the study area and its corresponding point cloud are illustrated in Fig. 1. In addition to the primary dataset, two supplementary datasets (Data 2 and Data 3) were employed to validate the model's generalizability. Both were acquired using the airborne LiDAR system developed under our research team's project, "Integration and Application of Aerial Remote Sensing Sensors" (hereafter referred to as the ARSSIA project). The specific system parameters are detailed in Table I.

This method is composed of four primary modules, it first employs a boundary extraction algorithm to obtain a rough coastal boundary, which is then used to re-extract the coastal point cloud, effectively eliminating a large number of irrelevant land points. Following the established correlation between multiple-echo laser returns (≥ 2 echoes) and vegetation signatures, analyze whether there are vegetation points in the coastal point cloud dataset. Detected vegetation points were

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TABLE I
SPECIFICATIONS OF THE DATASETS EMPLOYED IN THIS STUDY

data category	first collection		update time		laser sensor model	pulse frequency	scanning field of view angle	source
	year	month	year	month				
main data	2008	10	2023	6	ALS60	160,000HZ	75°	ARSSIA
data2	2008	10	2023	6	ALS60	160,000HZ	75°	ARSSIA
data3	2008	10	2023	6	ALS60	160,000HZ	75°	ARSSIA

subsequently processed through our standardized vegetation gap interpolation protocol, maintaining homogeneous point distribution and attribute consistency across both vegetated and nonvegetated surfaces. By analyzing the laser echo intensity values of coastal features, e.g., sand, gravel, soil, and rocks, threshold values are established to classify coastal point clouds into land-side and water-side subsets. Subsequently, a precise boundary extraction algorithm is applied to delineate: 1) the land-side boundary (deviating toward the water direction) and 2) the water-side boundary (deviating toward the land direction). These two extracted boundary point clouds are then merged to form a composite coastal boundary. Since the laser echo intensity values of submerged coastal features (e.g., sand, gravel, and soil) under shallow-water bodies are similar to those of fully exposed or on land, the merged boundary may deviate from the true water–land boundary and erroneously fall within the water area. To improve accuracy, the merged boundary is adjusted by shifting it landward by one average point spacing (the average point spacing, a fundamental parameter in the re-extraction of sea-area point clouds, often falls short of the distance between the merged boundary and the true water–land interface. This adjustment aimed at ensuring the merged boundary more accurately reflects the actual division). Based on the local terrain and elevation changes on the left and right sides of the water–land boundary line, design constraints for merging point cloud point attribute judgments, and screen the fusion point cluster into two types of boundary points with different attributes, namely, land-side boundary points and water-side boundary points. Afterward, the positions of these two types of boundary points will be optimized according to different attributes, and then fused and combined with tidal data and transformation models to convert them into coastal point clouds. The pre-extracted results of this study are corrected, adjusted, and fused using the extracted results of the coastal profile method (CPM), a currently superior extraction approach. Finally, a curve-fitting-based interpolation completion algorithm is applied to refine the fused shoreline, enhancing its smoothness and continuity. The overall workflow of the proposed method is illustrated in Fig. 2.

B. Coastal Zone Point Cloud Re-Extraction

Airborne LiDAR scanning of coastal zones generates extensive point cloud data encompassing both terrestrial locations and shallow subsurface features. However, for shoreline extraction applications, only points within a defined buffer zone surrounding the water–land boundary contain relevant information. The remaining point clouds not only lack utility for shoreline delineation but may also introduce unnecessary computational burdens, including storage inefficiencies, algorithmic complexity, and processing overhead. Point cloud re-extraction in coastal zones represents a critical preprocessing step that simultaneously satisfies shoreline extraction requirements and optimizes data volume. The airborne LiDAR point cloud not only has high-precision 3-D coordinate information, but also rich information on laser echo intensity, echo frequency, and color RGB values, which provide a foundation for the wide application of airborne LiDAR. To apply point cloud data in calculations, we define the following notations: let P_{text} represent a point cloud file named "text," $P_{text}(i)$ denote the ith point in the file, $P_{text}(i)_x, P_{text}(i)_y, P_{text}(i)_z$ represents the 3-D coordinates of point $P_{text}(i)$, $P_{text}(i)_I$ is its echo intensity value, and $P_{text}(i)_N$ is its echo count value.

When airborne LiDAR scans a coastal area, the flight path typically follows the shoreline, resulting in strip-shaped point cloud data. In the direction approximately perpendicular to the strip, the x- or y-coordinates of the point cloud exhibit a distinct variation pattern: they will repeat from small to large and then from large to small (or from large to small and then from small to large), forming a cyclical fluctuation. This characteristic can be leveraged to extract the approximate boundary of the strip point cloud facing the water body. Using this preliminary boundary, distance and elevation constraints can be applied to refine the re-extraction of the original sea point cloud based on the specified thresholds. Let $P_{original}$ denote the original coastal point cloud, P_{border} is the coarse boundary point cloud, and $P_{coastal}$ is the re-extracted point cloud. The computation proceeds as follows:

$$\begin{cases} P_{coastal} = \left\{P_{original}(j)\right\}, \Delta d \le D \& \Delta h \le H \\ \Delta d = d\left(P_{original}(j), P_{border}(i)\right) \\ \Delta h = \left|P_{original}(j)_z - P_{border}(i)_z\right| \end{cases} \tag{1}$$

where Δd denotes the distance between two points $P_{original}(j), P_{border}(i)$, Δh represents their elevation difference, and D and H are the predefined thresholds for distance and elevation difference, respectively.

C. Vegetation Detection and Filling (VDF) Module: Filling Gaps Under Vegetation Cover (If Vegetation Point Clouds Exist)

The shoreline is defined as the boundary formed by the intersection of the land surface and the mean high-tide level. Shorelines can be categorized into natural and artificial types:

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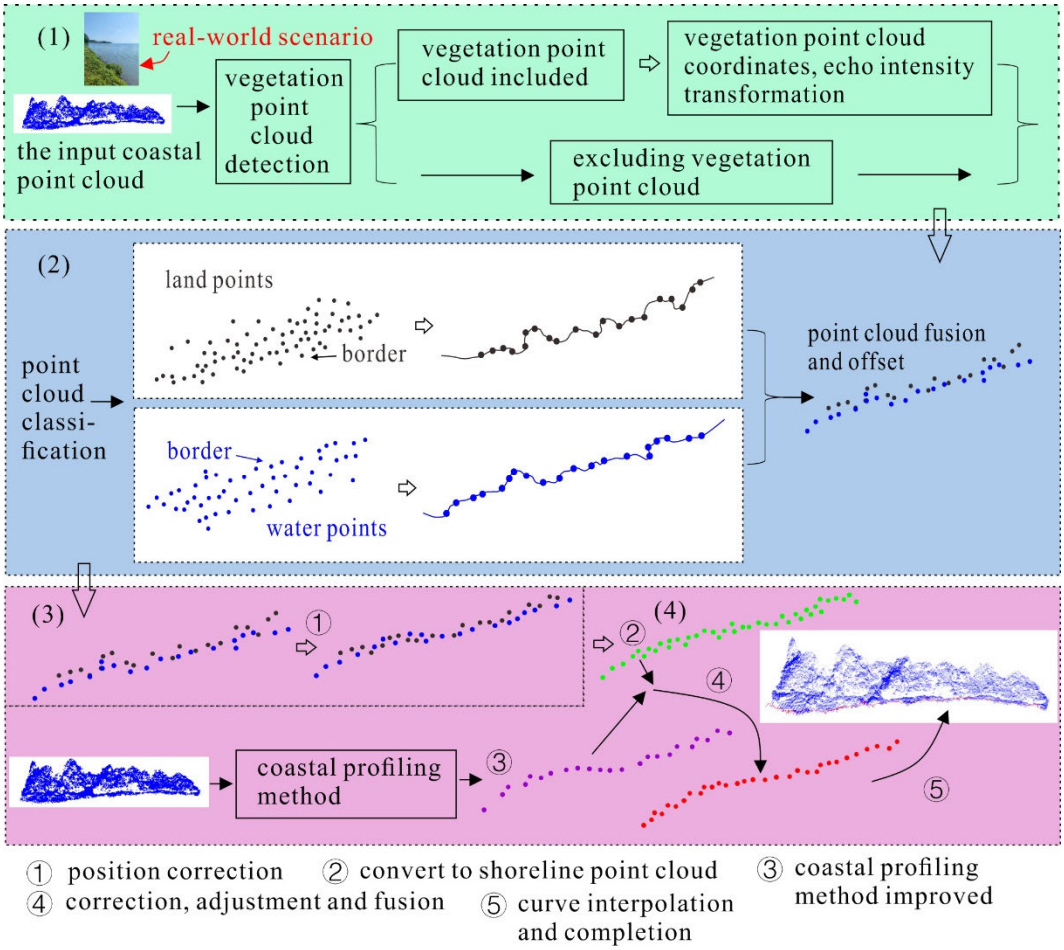


Fig. 2. Workflow of the method, Shoreline_All.

natural coasts include shallow waters, muddy coasts, vegetated coasts, and bedrock coasts, artificial coasts encompass urban residential land, ports/docks, and other human-made infrastructure along the shoreline. For most coastal types, airborne LiDAR can directly capture accurate geospatial information about the coastal zone. However, vegetated coasts present a unique challenge: dense foliage blocks a significant portion of the laser pulses, preventing them from reaching the underlying terrain. This results in data gaps in the point cloud, particularly in areas beneath vegetation cover where the laser beam cannot penetrate. To accurately capture water and land conditions beneath vegetation, it is necessary to convert the point clouds on the branches and leaves to the ground according to a certain method. The proposed method preserves the horizontal (x, y) coordinates of these points while projecting their elevation (z-coordinate) downward. The projected elevation is then assigned the average elevation value of the k-nearest neighboring ground points, as illustrated in Fig. 3.

Due to potential severe data loss in coastal areas obscured by vegetation, the algorithm adaptively increases the neighborhood order (k) by increments of 1 until at least two points are detected within the kth order neighborhood. Let the point cloud on coastal branches and leaves be $\mathrm{P_{veget}}$ and its classification method is as follows:

$$\mathrm{P_{veget}} = \{\mathrm{P_{coastal}}(j)\}, \mathrm{P_{coastal}}(j)_N \geq 2. \tag{2}$$

Let the point set within the k-order neighborhood be $\mathrm{P_k}$. Then, after projecting the vegetation point cloud set, the elevation of each point is

$$h_j = \frac{\sum_{i=1}^{n} \mathrm{P_k}(i)_z}{n}. \tag{3}$$

Then, the projected coordinates of each vegetation point are

$$\begin{aligned} x_j &= \mathrm{P_{veget}}(j)_x \\ y_j &= \mathrm{P_{veget}}(j)_y \\ z_j &= h_j. \end{aligned} \tag{4}$$

Vegetation-covered sea areas represent a distinct coastal terrain type where remote sensing data in subsurface spaces are often sparse or unavailable due to obstruction by vegetative canopies. The methodology presented in Section II-C enables spatial data interpolation within these vegetation-occluded zones. For point cloud classification based on laser echo intensity values, intensity assignment for the interpolated points must be systematically addressed. The proposed approach follows a similar scheme to (3). For each interpolated point, the laser echo intensity value is assigned as the mean intensity of all points within its k-order neighborhood. This ensures that every point in the point cloud dataset $\mathrm{P'_{coastal}}$ contains both

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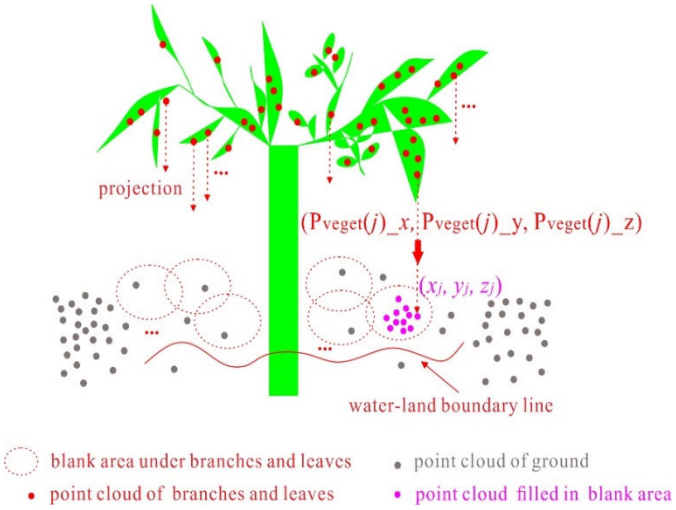


Fig. 3. Spatial information interpolation under branches and leaves.

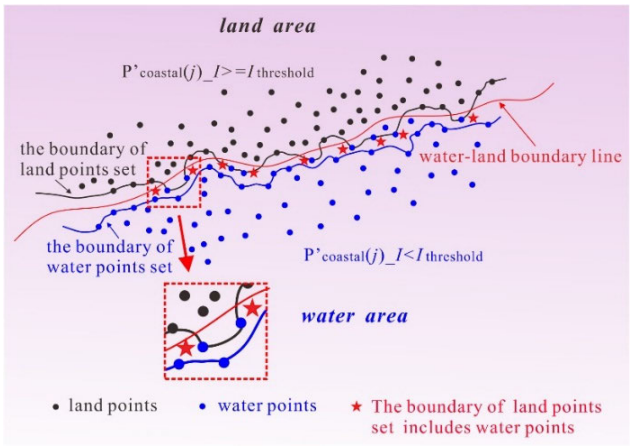


Fig. 4. Classification of land-side and water-side point clouds and precise boundary extraction.

3-D coordinates and calibrated intensity values. The intensity assigned to an interpolated point is computed as

$$I_j = \frac{\sum_{i=1}^{n} \mathrm{P_k}(i)_I}{n}. \tag{5}$$

The projected point set is merged with the remaining point set after $\mathrm{P_{veget}}$ subtraction in $\mathrm{P_{coastal}}$ forming the newly filled marine point set $\mathrm{P'_{coastal}}$. For nonvegetation marine point clouds, $\mathrm{P'_{coastal}}$ is equivalent to $\mathrm{P'_{coastal}}$.

D. Classification, Extraction, and Processing (CEP) Module: Marine Point Cloud Classification, Precise Boundary Extraction

Different land cover types exhibit distinct laser reflection intensities due to their varying surface properties. Researchers have systematically calibrated the laser echo intensity values for diverse land cover categories. These calibrations account for standard operational conditions, including typical flight altitudes and common emission pulse intensities on airborne platforms, thereby establishing reference intensity ranges for prevalent land cover types. For the point cloud set $\mathrm{P'_{coastal}}$, it consists of points located on both sides of the water–land boundary during data acquisition, namely, the land-side and water-side point clouds. When nonbathymetric LiDAR scans from the land side toward the water body, the quantity of reflected point clouds decreases progressively with increasing water depth. At a certain depth threshold, the laser beams become completely absorbed by the water with no return signals. Concurrently, the intensity of reflected laser points gradually diminishes as the scanning proceeds into deeper waters. The laser echo signals in water areas primarily originate from submerged sand, gravel, and soil. The echo intensities of these underwater features in shallow regions may be the same or similar to those of their terrestrial counterparts. As water depth increases, the intensity values progressively decrease until the signal is completely attenuated. Based on this phenomenon, the intensity threshold for distinguishing water areas can be set slightly below the calibrated values (test results show that setting a values 1–3 below the calibration threshold positions the extracted boundary within the water area and maximally close to the true water–land interface) for sand, gravel, and soil in typical land environments. However, near the water–land boundary, submerged features within a short distance may exhibit intensity values comparable to those on land. Consequently, land-side point clouds classified using this threshold often include both terrestrial points and shallow underwater sand, gravel, and soil points. Airborne LiDAR point clouds exhibit varying spacing, due to factors such as flight altitude and laser spot size. Consequently, the two classified point cloud types by this threshold—land-side points and water-side points—do not share identical boundaries (land-side point clouds biased toward the water direction boundary and water-side point clouds biased toward the land direction boundary). Subsequently, we propose refining the rough boundary extraction algorithm to derive precise boundaries for each cluster: by fusing these two precise boundaries, we obtain a more accurate representation of the water–land boundary, as illustrated in Fig. 4.

Point cloud classification process

$$\begin{cases} \mathrm{P_{land}} = \left\{\mathrm{P'_{coastal}}(j)\right\}, \mathrm{P'_{coastal}}(j)_I \geq I_{\mathrm{threshold}} \\ \mathrm{P_{water}} = \left\{\mathrm{P'_{coastal}}(j)\right\}, \mathrm{P'_{coastal}}(j)_I < I_{\mathrm{threshold}} \end{cases} \tag{6}$$

where $\mathrm{P_{land}}$ and $\mathrm{P_{water}}$ denote the land-side and water-side point cloud datasets, respectively, and $I_{\mathrm{threshold}}$ represents the intensity threshold.

Accurate boundary extraction algorithm: 1) partitioning: divide the point clusters on both land and water sides along the shoreline into blocks based on x- or y-coordinates, with a predefined interval for each block; 2) boundary extraction: apply the boundary extraction method described in Section II-B to identify points with extreme x or y values within each block. These points are designated as the block boundary points; and 3) fusion: merge the boundary points from all blocks to reconstruct the precise boundary point cloud of the entire cluster.

Let $\mathrm{P_{land_b}}$ and $\mathrm{P_{water_b}}$ denote the precise boundary point clouds extracted from the point cloud sets $\mathrm{P_{land}}$ and $\mathrm{P_{water}}$, respectively. The fused point cloud set $\mathrm{P_{border_1}}$ is obtained by merging $\mathrm{P_{land_b}}$ and $\mathrm{P_{water_b}}$. As previously demonstrated,

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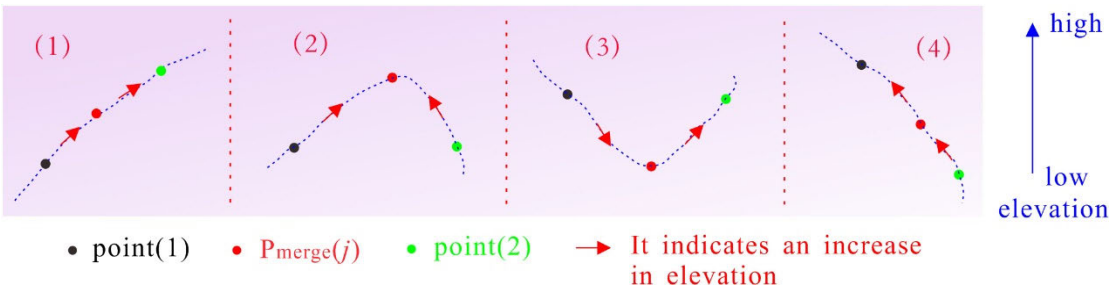


Fig. 5. Four elevation variation relationships between point (1), point (2), and point $P_{merge}(j)$.

P_{border_1} tends to deviate from the actual land–water boundary and shift toward the water side. To compensate for this bias, each point in P_{border_1} is offset toward the land side by one times the average point spacing P_{border_1} of $P'_{coastal}$, resulting in the corrected point cloud set P_{merge}.

E. Precision Transition (PT) Module: Position Refinement

1) Attribute Judgment of Points in P_{Merge}: The point position in the point cloud set P_{merge} can roughly represent the shape of the land–water boundary. To enhance the accuracy when transforming these boundary points into shoreline points through a shoreline conversion model, it is necessary to optimize the positional precision of each point in P_{merge}. Water flow typically follows the topographic gradient, moving from higher to lower elevations. In certain regions near the land–water boundary (on both the terrestrial and aquatic sides), there will be a transition from high terrain to low terrain. The points from point cloud set P_{merge} are repositioned into the re-extracted coastal zone point cloud set $P'_{coastal}$. For each target point $P_{merge}(j)$ in this set P_{merge}, screen for the nearest neighboring points on both sides and analyze the elevation variation between these two adjacent points and the target point $P_{merge}(j)$ to assess local terrain transitions. The classification of each point in $P_{merge}(j)$ as land or water point is determined by analyzing the surrounding terrain conditions. Subsequently, coordinate optimization is performed on a per-point basis, according to the principle that the closer the point to the actual water–land boundary, the higher the accuracy.

For shorelines with an east–west orientation (either east-to-west or west-to-east), we compute points in the upper and lower spatial regions relative to the central point $P_{merge}(j)$. For north–south oriented shorelines (north-to-south or south-to-north), we instead calculate points in the left and right regions (in the case of long-distance, large-scale coastlines, segmentation is feasible based on the criterion of continuous azimuth change between adjacent boundary points over a given coastline length. The coastline can thereby be divided into a series of segments, predominantly oriented east–west or north–south, for the purpose of independent subsequent processing). The following steps demonstrate the process for filtering points in the upper region:

$$P_{up} = \left\{P_{up}(i)\right\}, P_{up}(i) \in P'_{coastal}. \tag{7}$$

The constraint conditions are

$$\begin{cases} \theta\left\{\mathbb{R}\left(P_{up}(i), P_{merge}(j)\right), \left[\mathbb{R}\perp\mathbb{R}\left(P'_{coastal}\right)\right]\right\} \le \Delta\theta \\ P_{up}(i)_y > P_{merge}(j)_y \end{cases} \tag{8}$$

where $\mathbb{R}(P'_{coastal})$ represents the shoreline direction, $\mathbb{R}\perp\mathbb{R}(P'_{coastal})$ denotes the direction perpendicular to the shoreline, $\mathbb{R}(P_{up}(i), P_{merge}(j))$ indicates the connecting line direction between two points, θ is the angle between the connecting line direction and the perpendicular direction of the shoreline, and $\Delta\theta$ is the threshold for θ(typically set to 30°–45°). Similarly, this method can be used to filter the point sets for the lower spatial region or the left/right spatial regions.

For a given point in the fused point cloud set P_{merge}, we can filter the corresponding constrained regional point sets for its upper, lower, left, or right parts. Then, within these constrained point sets, we search for the point with the closest planar distance to point $P_{merge}(j)$. Taking the shoreline's primary orientation (south-to-north or north-to-south) as an example, we can filter the upper and lower constrained point sets P_{up} and P_{down} for each point $P_{merge}(j)$. Within these sets, the nearest planar-distance points to $P_{merge}(j)$ are denoted as point(1) and point(2), respectively. The elevations of point(1) and point(2) relative to point $P_{merge}(j)$ exhibit four possible variation relationships: 1) Z(point(1)) <= $P_{merge}(j)_z$ <= Z(point(2)); 2) Z(point(1)) <= $P_{merge}(j)_z$ >= Z(point(2)); 3) Z(point(1)) >= $P_{merge}(j)_z$ <= Z(point(2)); and 4) Z(point(1)) >= $P_{merge}(j)_z$ >= Z(point(2)); as shown in Fig. 5.

According to the principle of water flow direction from high to low elevation, the local terrain elevation on both sides of the water–land boundary should exhibit a descending topographic profile. For the three points—point(1), point(2), and reference point $P_{merge}(j)$—the elevation relationships determine their attribute classification, when satisfying conditions (1)–(3), point $P_{merge}(j)$ is classified as the land-side point, when condition (4) is met, point $P_{merge}(j)$ is designated as the water-side boundary point. Notably, case (4) may theoretically occur with all three points on the landside. However, in such scenarios: point $P_{merge}(j)$ approximates the crude water–land boundary and point $P_{merge}(j)$ inherently tends toward boundary adjacency. If point(2) were actually a land-side point at a significant distance from the boundary, this would contradict $P_{merge}(j)$'s role as a boundary indicator. Therefore, case (4)

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conclusively classifies $P_{merge}(j)$ as the water-side boundary point regardless of theoretical possibilities.

2) Point Position Optimization Based on Point Cloud Attribute: Due to point spacing in airborne LiDAR point clouds, the fused boundary point set P_{merge} can only approximate the water–land boundary. Using the above method, points in set $P_{merge}(j)$ can be classified as either land-side boundary points or water-side boundary points. Both types of boundary points are then offset by appropriate distances to better align with the actual water–land boundary. Since the exact offset distance cannot be determined from current data conditions, we define it as half of the average point spacing. The specific offset distances are.

When the shoreline is above the land

$$\begin{aligned} P_{merge}(j)_x &= P_{merge}(j)_x \mp \frac{1}{2}\sigma * \cos\alpha \\ P_{merge}(j)_y &= P_{merge}(j)_y \pm \frac{1}{2}\sigma * \sin\alpha \\ P_{merge}(j)_z &= P_{merge}(j)_z. \end{aligned} \tag{9}$$

In the above equation, the boundary points on the land-ward side are assigned negative (−) and positive (+) value for the corrected x- and y-coordinates, respectively. Conversely, the boundary points on the waterside are assigned positive (+) and negative (−) value for the corrected x- and y-coordinates, respectively. When the shoreline is situated below the landmass

$$\begin{aligned} P_{merge}(j)_x &= P_{merge}(j)_x \pm \frac{1}{2}\sigma * \cos\alpha \\ P_{merge}(j)_y &= P_{merge}(j)_y \mp \frac{1}{2}\sigma * \sin\alpha \\ P_{merge}(j)_z &= P_{merge}(j)_z. \end{aligned} \tag{10}$$

In the above equation, the boundary points on the landward side are assigned positive (+) and negative (−) value for the corrected x- and y-coordinates, respectively. Conversely, the boundary points on the waterside are assigned negative (−) and positive (+) value for the corrected x- and y-coordinates. In (9) and (10): σ represents the average point spacing, α denotes the angle between the horizontal axis and the direction perpendicular to the shoreline trend. This optimization process refines the positions of points in set P_{merge}, enabling P_{merge} to more accurately represent the true water–land boundary.

F. Collaboration and Smoothing (CAS) Module: Strict Shoreline Generation, CPM Fusion, and Shoreline Fitting

1) Shoreline Conversion Model: Three primary models are used to convert the water–land boundary into the average high-tide shoreline elevation: traditional linear tidal level correction model, linear fitting tidal level correction model, and plane profile-based tidal level correction model. Historically, the traditional linear tidal correction model has been applied to transform water–land boundaries into strictly defined shorelines. However, this method relies on two preexisting water–land boundary lines to compute the coastal slope, introducing potential errors due to inaccuracies in their initial positioning. These errors propagate, reducing the accuracy of the corrected distance. To address this limitation, this study adopts the linear fitting tidal level correction model for shoreline conversion. The key steps include: measuring multiple coastal slope points using real-time kinematic (RTK) technology to determine the slope angle (θ). Combining this with the annual average high-tide line position and the instantaneous water–land boundary to compute the correction distance (d) required for adjustment (see Fig. 6). This approach improves accuracy by reducing dependency on error-prone boundary lines and leveraging direct RTK-measured slope data.

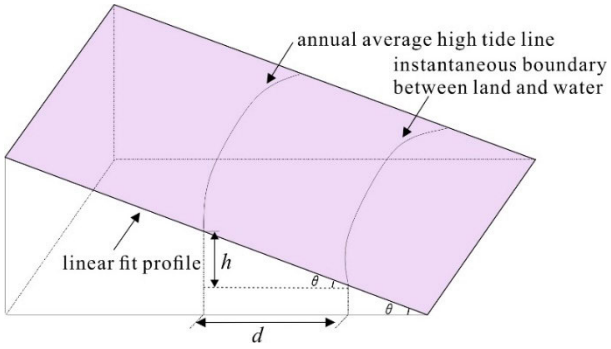


Fig. 6. Schematic of the linear fitting tidal level correction model.

2) Correction, Adjustment, and Fusion of Results Using the Joint CPM: The CPM and contour tracking method (CTM) derive shoreline point clouds by applying mathematical transformations to coastal elevation data based on the average high-tide line elevation in the study area. Alternatively, another approach involves first extracting the water–land boundary line and subsequently converting it into a strictly defined shoreline through a transformation model. This study focuses on the latter method (i.e., transforming water–land boundaries into shorelines) but further integrates the CPM—a direct shoreline calculation technique—to enhance the accuracy and density of shoreline point clouds through correction, adjustment, and fusion. Both the CPM and CTM rely on long-term average high-tide elevation data. The CPM defines the shoreline as the outermost intersection line between the coastal terrain and a planar surface at this elevation. However, this method may yield errors in two scenarios: the original coastal data fail to fully represent the terrain due to obstructions (e.g., clouds, topography, or data collection artifacts), resulting in missing coastal segments. The physical shoreline is obscured by transient objects (e.g., debris or vegetation), leading to incorrect peripheral delineation. These errors manifest differently in the extracted point clouds (see Fig. 7): when there is only the first scenario, the incorrect shoreline point cloud is shown in Fig. 7 (1), and when there is only the second scenario, it is shown in Fig. 7 (2). Both scenarios exist and there is a correct shoreline point cloud in between, as shown in Fig. 7 (3). When both scenarios exist and are adjacent, it is shown in Fig. 7 (4). Normal shoreline point clouds exhibit small interpoint distances (d), whereas erroneous clusters (protrusions or recessions) deviate by a larger, variable distance (D). Existing CPM lack mechanisms to address such errors. To mitigate this, we propose an algorithm to identify and remove erroneous point clouds (based on the patterns in Fig. 7), followed by fusion with shoreline point clouds generated by

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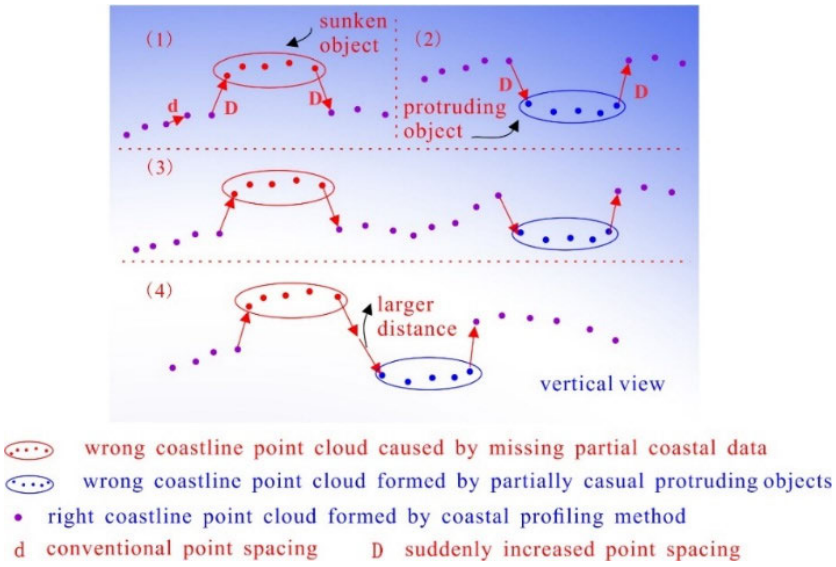


Fig. 7. Error point detection and removal via CPM.

our primary method. This integration increases the point cloud density, enabling finer-scale shoreline characterization.

In addition, a 3-D distance threshold is applied to fused point clouds: if adjacent points fall within the threshold, their midpoint replaces the original pair, if distances or elevation differences exceed the threshold, manual intervention is required to retain the most plausible points.

By correcting and adjusting results through the joint CPM, the fused shoreline point clouds achieve enhanced detail resolution, yielding a more natural and smooth shoreline morphology.

3) Shoreline Fitting Interpolation Completion: The shoreline point cloud generated in Section II-E exhibits nonuniform point spacing. Directly connecting these points may result in unnatural geometric features, including abrupt protrusions or concavities. To achieve a smooth and continuous shoreline that accurately represents the actual coastal morphology, we implement a curve fitting and interpolation process with the following criteria: interpolation rules: ① distance <0.5 m: no interpolation (zero additional points), ② 0.5 m ≤ distance <1 m: insert 2 points, ③ 1 m ≤ distance <1.5 m: insert 3 points, ④ 1.5 m ≤ distance <2 m: insert 4 points, and ⑤ distance ≥ 2 m: process the first 2-m segment according to the 1.5–2-m rule, then recursively apply the appropriate rule to the remaining segment until the residual distance is less than 0.5 m. Each interpolated point is constrained to lie within the local curvature trend of the shoreline. The algorithm workflow is illustrated in Fig. 8.

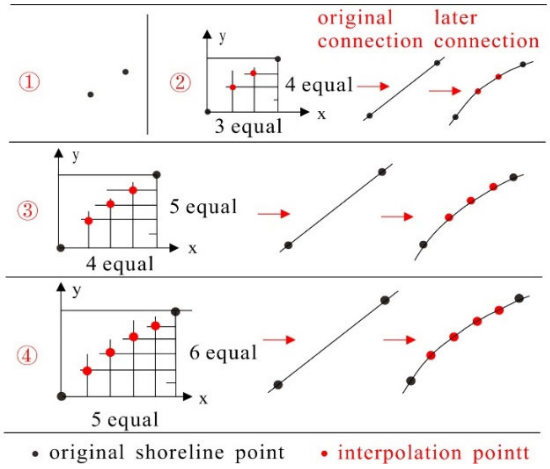


Fig. 8. Principle of shoreline fitting, interpolation completion.

G. Accuracy Evaluation Indicators

Currently, there is no universally applicable method for evaluating the accuracy of shorelines extracted from point cloud data. In theory, if field mapping is feasible, the extracted shoreline should be compared with ground-truth measurements. However, for large-scale or diverse marine environments, most coastal areas are inaccessible, making conventional field surveys impractical. To address this challenge, an alternative approach involves generating multiple shoreline datasets as observational samples. Based on error theory, the mean observation (i.e., the average shoreline position) can be derived from these samples. By comparing each set of observations with the mean value, the accuracy of individual datasets—and thus the overall accuracy of each extraction method—can be quantified. For further details, refer to [22].

III. Experiments and Analysis

A. Shoreline Extraction Using the Proposed Method

The proposed framework is applicable to all coastal types. When vegetation point clouds are present, the method reconstructs the underlying spatial information obscured by vegetation. In cases where no vegetation point clouds exist, this step is automatically skipped. To validate the robustness of our approach, we selected extensive datasets covering long shorelines with complex terrain and diverse shoreline types. The selected datasets demonstrate strong representativeness for validating our proposed methodology. In addition, the

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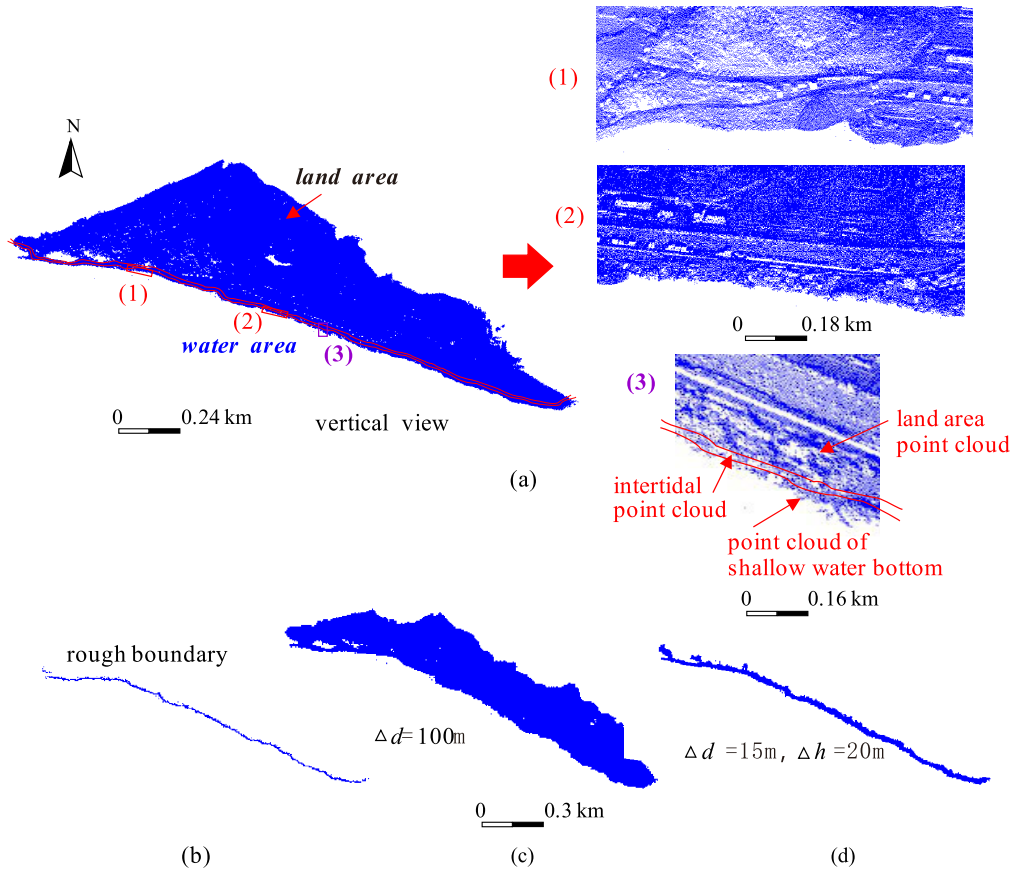


Fig. 9. Coastal zone point cloud re-extraction. (a) Original coastal point cloud. (b) Rough boundary extraction. (c) Refined coastal point cloud ($P_{coastal_1}$) using a 100-m plane distance threshold from the boundary. (d) Further refined coastal point cloud ($P_{coastal_2}$) with a 15-m distance threshold and 20-m height difference threshold.

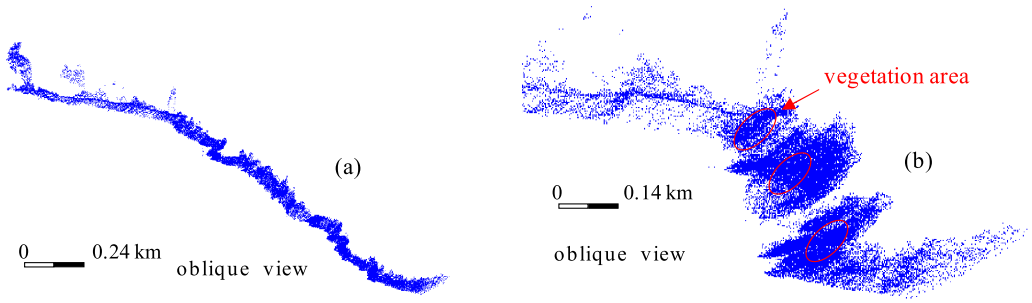


Fig. 10. Enlarged view of $P_{coastal_2}$. (a) Overall enlarged view. (b) Vegetation coast area.

offshore water area contains multiple clustered noise points with varying aggregation levels. These spurious point clouds are detected and removed based on their disconnection from the primary landmass cluster, as illustrated in Fig. 9(a). To clarify the distribution of point clouds across various coastal terrains, the main dataset was annotated to distinguish between sea and land areas. Two representative transects with approximate spatiotemporal positions were drawn in the seaward sector for illustrative purposes. Furthermore, as shown in the enlargement of the orange box in 3rd, the point clouds were classified into shallow-water subsurface features, intertidal zones, and land areas. For the subsequent Data 2 and Data 3, only sea and land labels were applied, as the spatial distribution of terrain point clouds is similar to that of the main dataset. The experimental data were imported into existing point cloud processing software using a spatial 3-D coordinate system, where the x-axis points to the right, the y-axis forward, and the z-axis vertically upward. The results show that the shoreline runs approximately east–west (or west–east) and is situated above the land. By analyzing the coordinate trends, a rough boundary was extracted, as illustrated in Fig. 9(b). Using this rough boundary point cloud as a reference, *a* plane distance constraint of Δd = 100 m from the boundary was applied to refine the

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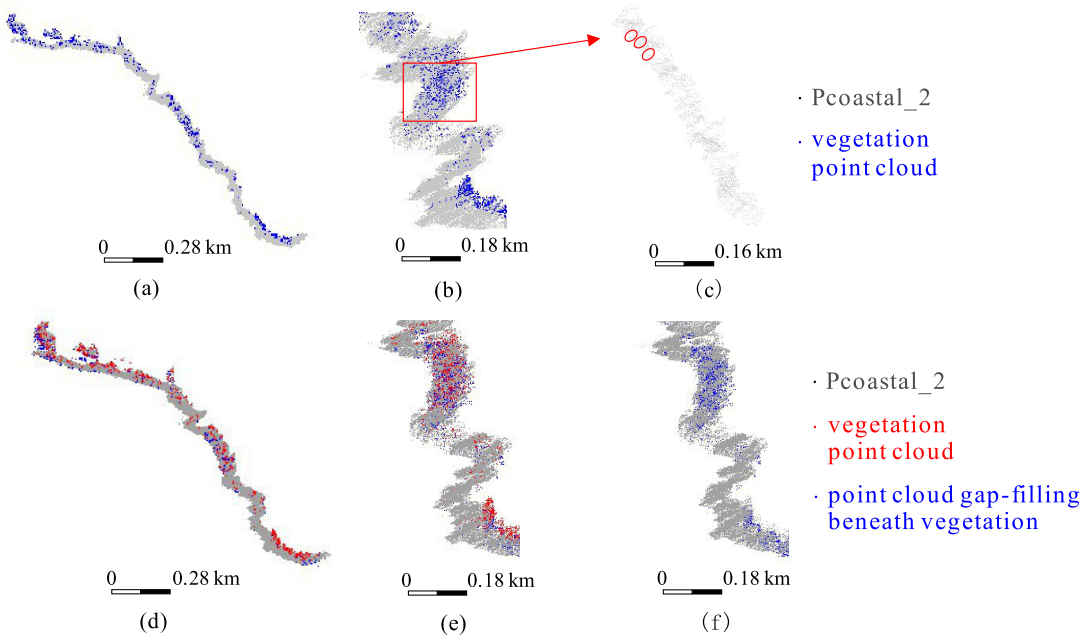


Fig. 11. Vegetation point cloud detection and filling process: (a) vegetation point cloud display, (b) selecting areas with dense vegetation, (c) removing vegetation point clouds from the boxes in the dense vegetation area map, (d) overlaying vegetation points, filling points with sea-area point clouds, (e) corresponding to (b) and adding filling points, (f) corresponding to (e) and removing original vegetation points (retaining only filling points).

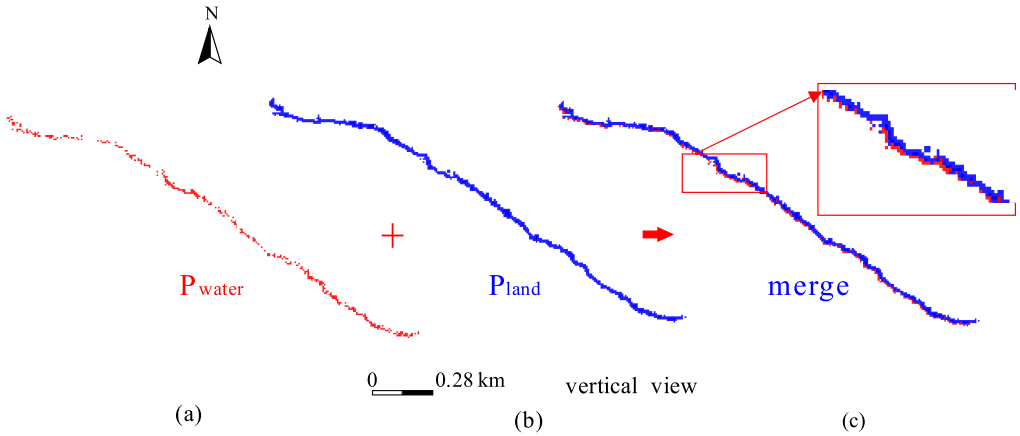


Fig. 12. (a) Water-side point cluster, (b) land-side point cluster, and (c) superposition of land and water-side point cluster.

coastal zone point cloud extraction. The final re-extracted coastal zone point cloud is presented in Fig. 9(c). To facilitate future classification of sea-area point clouds based on laser echo intensity—while minimizing misclassification of land points that meet the intensity threshold—the re-extraction constraints were adjusted to Δd = 15 m and Δh = 20 m. The refined coastal point cloud extraction results are shown in Fig. 9(d). These two re-extracted coastal point cloud datasets are hereafter referred to as $P_{coastal_1}$ and $P_{coastal_2}$.

To better visualize the detailed structure of the point cloud cluster, an enlarged view is presented at a larger scale in Fig. 10(a). The analysis reveals the presence of multiple coastal types within the cluster. Specifically, vegetated coastal areas were identified with their point cloud distribution displayed from a side perspective in Fig. 10(b).

Following the extraction of the coastal point clouds, the vegetation point cloud detection algorithm was applied to classify the vegetation, as shown in Fig. 11(a). A densely vegetated section was selected and enlarged in Fig. 11(b), revealing a substantial number of vegetation points above the ground. The area within the box in Fig. 11(b) was further enlarged, and the vegetation points were filtered out to produce Fig. 11(c). The small ellipse in this panel highlights a representative data gap, illustrating the extensive information loss caused by the canopy. This gap was then processed using our proposed spatial data filling algorithm for areas under vegetation, resulting in Fig. 11(d). In this visualization, red indicates the original vegetation points, while blue represents the newly filled points (for clarity, the original color of vegetation points has been changed to red in this panel). Fig. 11(e) shows the original image corresponding to the

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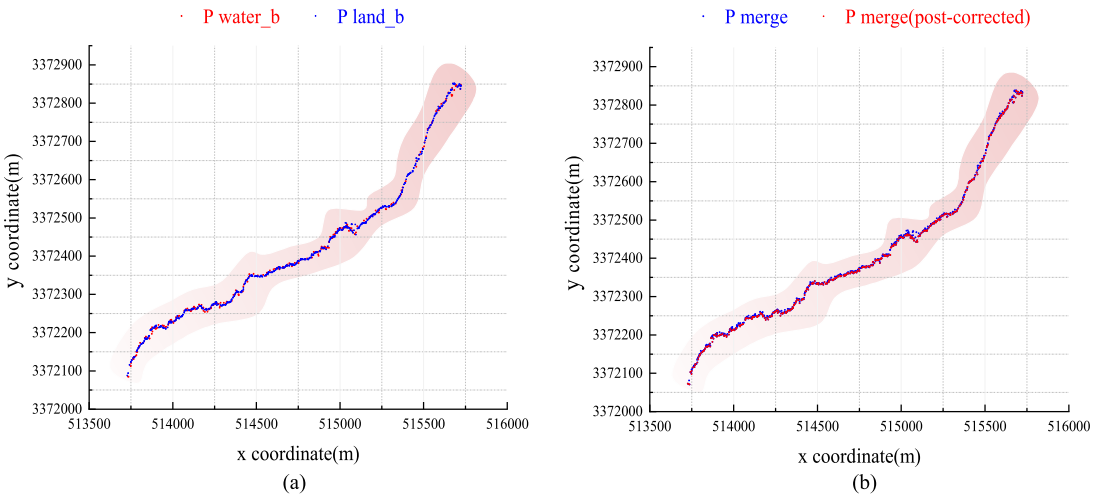


Fig. 13. (a) Superposition of P_{water_b} and P_{land_b} and (b) superposition of P_{merge} and corrected P_{merge}.

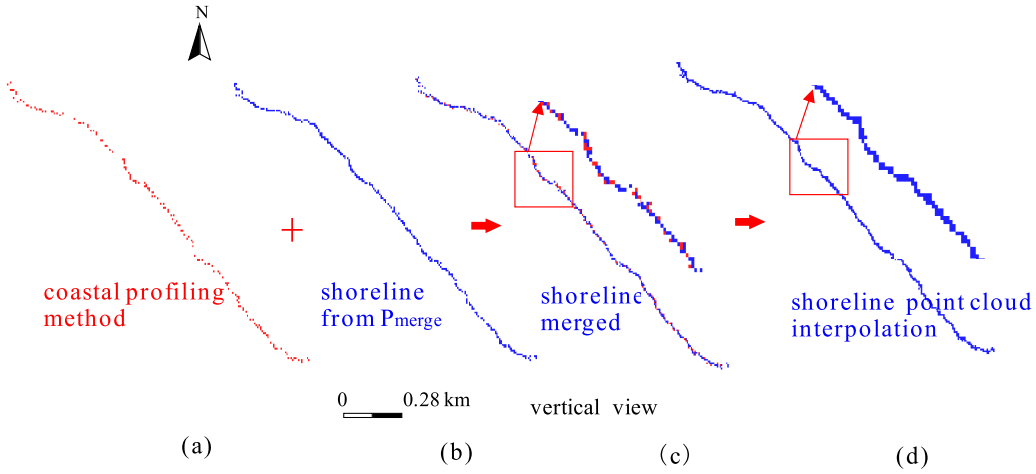


Fig. 14. (a) Shoreline point cloud corrected by CPM, (b) shoreline point cloud converted from P_{merge}, (c) corrected, adjusted, and fused results of (a) and (b), and (d) completion results through curve-fitting interpolation method.

scene in Fig. 11(b) and (f) presents the same scene after the vegetation point cloud has been removed and the underlying space has been filled.

Thus, Fig. 11(b) and (f) collectively demonstrate the point cloud before and after the vegetation filling process.

The vegetation point cloud clusters within the coastal re-extraction dataset were processed using the method outlined in Section II-C. This process supplemented ground spatial information obscured by vegetation branches and leaves, assuming that the new coastal point cloud cluster remains $P_{coastal_2}$. Based on statistical analysis of laser echo intensities for sand, gravel, and soil, a threshold value slightly below the minimum observed intensity, approximately 31 was selected. Using this intensity threshold, the point clouds were classified into water-side point clusters [see Fig. 12(a)] and land-side point clusters [see Fig. 12(b)]. The two classified clusters were then superimposed to generate the composite representation in Fig. 12(c).

Using a precise boundary extraction algorithm, the respective boundaries of the land-side and water-side point clouds are extracted as P_{land_b} and P_{water_b}. The two boundaries are then merged and offset land-ward by one times the average point spacing to obtain P_{merge}. Subsequently, following the method described in Section II-D, the boundary points belonging to land and water are filtered and corrected separately, then fused to obtain the corrected P_{merge}, as shown in Fig. 13(a) and (b), respectively.

The shoreline was calculated using the CPM, with initial removal of anomalous point clouds from protruding/recessed features [see Fig. 14(a)]. According to the shoreline conversion model, The conversion model assumes that the coast of this area is a slope with a constant inclination angle, combined with local hydrological and tidal data (which can be queried at the local hydrological station), to obtain an average inclination angle of 30.568° for the slope in this sea area. The annual average high-tide line elevation collected by the hydrological observation station is 1.9 m (China 1985 elevation standard), and then combined with is converted into a shoreline point cloud and shown in Fig. 14(b). Then, the two are fused to produce an integrated coastal boundary [see Fig. 14(c)]. Finally, the shoreline completion point cloud is obtained by a curve-fitting interpolation method in Section II-F and shown

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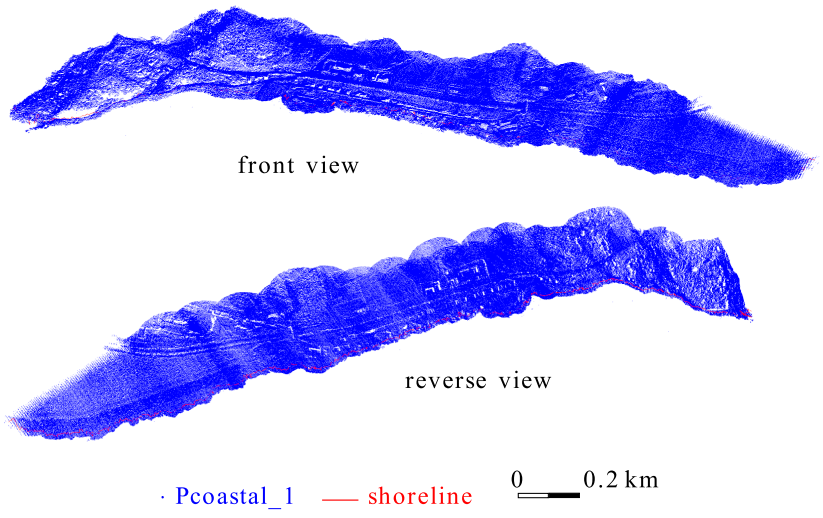


Fig. 15. Overlay of shoreline and coastal zone point clouds $P_{coastal_1}$.

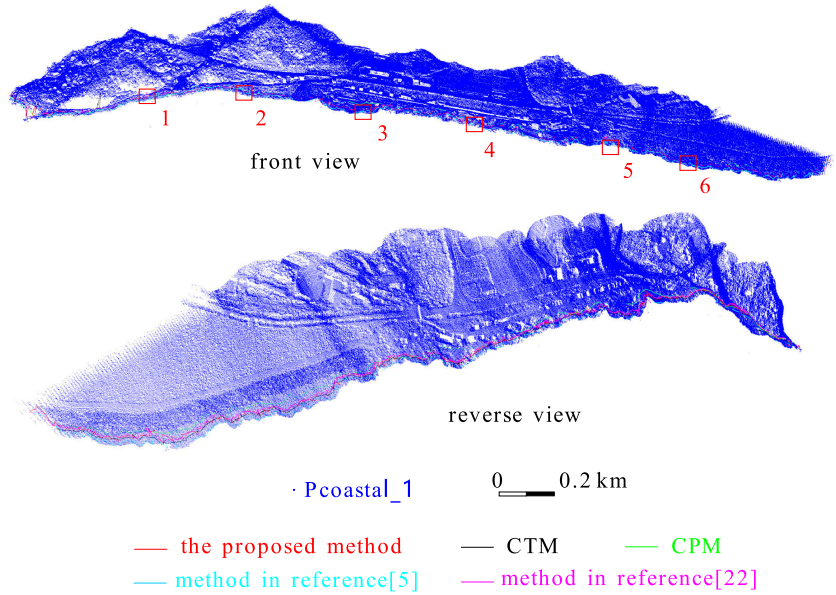


Fig. 16. Superposition of shoreline generated by the first five different methods and coastal zone point clouds.

in Fig. 14(d). P_{merge} integrating these parameters is converted into a shoreline point cloud and shown in Fig. 14(b). Then, the two are fused to produce an integrated coastal boundary [see Fig. 14(c)]. Finally, the shoreline completion point cloud is obtained by curve-fitting interpolation method in Section II-F and shown in Fig. 14(d).

After performing the interpolation, sort the shoreline point cloud data in ascending order based on their x-coordinates. For points with identical x-coordinates, sort them by y-coordinate in ascending order. If both x and y are the same, further sort them by z-coordinate in ascending order. Once sorted, sequentially connect the points to generate a smooth curve. Finally, overlay this curve with the re-extracted shoreline point cloud $P_{coastal_1}$, as illustrated in Fig. 15.

B. Comparison With Other Methods

The current mainstream methods for shoreline extraction from airborne LiDAR point clouds include CTM and CPM. To evaluate the performance of our proposed approach, we compare it with two recognized conventional techniques, two newer ones introduced in [5] and [22], as well as two advanced models, namely, U-Net and PointNet++. All seven methods were applied to extract and process the shoreline from the experimental dataset. Due to the limited color palette in the PCL visualization library, which restricts line colors to combinations of red, green, and blue (RGB) colors in combination with 0 and 255, the line colors can only be set to red, green, blue, cyan, pink, white, and black. Only seven distinct colors are available. The first five methods evaluated (including the proposed one) utilized these available colors, leaving no unique RGB combinations for the remaining two advanced methods (U-Net and PointNet++). Therefore, to ensure a clear and unambiguous visual comparison, this study employs a sequential evaluation strategy. First, the proposed method is compared against four other approaches (CTM, CPM, and the methods from [5] and [22]). Subsequently, a separate comparative analysis is conducted with U-Net and PointNet++. For a precise assessment of shoreline shape and

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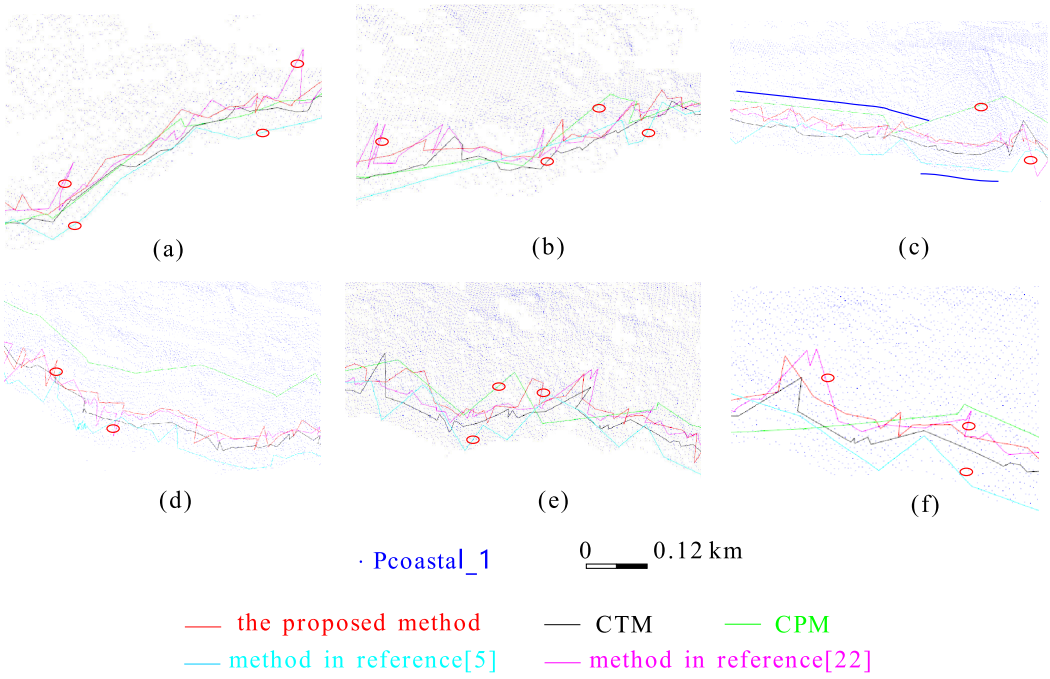


Fig. 17. Presents enlarged views of the regions marked in Fig. 16, (a) box 1, (b) box 2, (c) box 3, (d) box 4, (e) box 5, and (f) box 6.

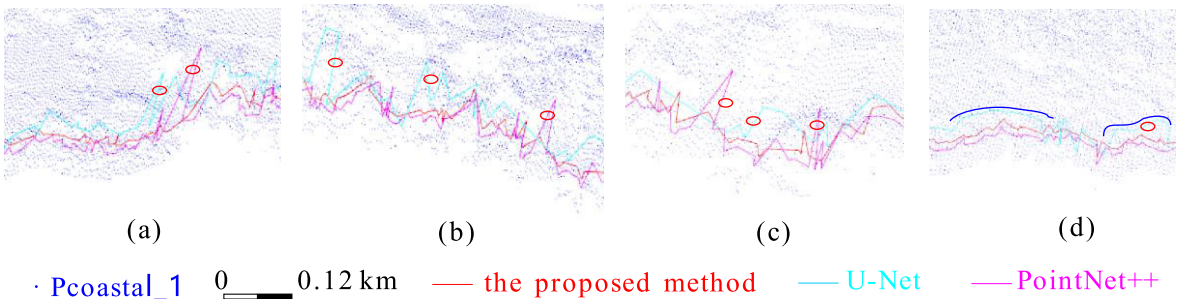


Fig. 18. Comparative results among the proposed method, U-Net, and PointNet++. (a)–(d) Enlarged views of four randomly selected areas.

positional accuracy, the results of the initial five methods (the proposed method, CTM, CPM, and the methods from [5] and [22], the same below) are overlaid onto the re-extracted sea-area point cloud, as illustrated in Fig. 16.

For a more detailed comparison, randomly selected regions of interest (boxes 1–6) in Fig. 16 were enlarged and displayed separately as Fig. 17 to better illustrate the differences between shoreline extraction methods.

U-Net and PointNet++ are two fundamentally distinct deep learning models that differ in their input modalities. For U-Net, we adopt a rasterization approach to convert irregular point clouds into regular 4-channel images. The network follows a classic symmetric encoder–decoder architecture with skip connections for feature fusion, contains four depth levels, and starts with 64 filters. It is trained using a combined Dice-cross-entropy loss and optimized with Adam. In contrast, PointNet++ directly processes multidimensional point sets. Its hierarchical learning structure comprises four set-abstraction layers, with multiscale grouping incorporated at the second and third layers to facilitate cross-scale feature fusion. To mitigate class imbalance, it is trained using weighted cross-entropy loss and optimized via SGD. The outputs from both models undergo a unified postprocessing pipeline to derive the final vector coastlines. All models are trained on the same coastline-extraction dataset established in our prior work, with hyperparameters tuned on a common validation set. Performance is ultimately evaluated and compared on the independent test set presented in this article.

Four representative areas were randomly selected from the comparison chart between the proposed method and the advanced U-Net and PointNet++ models, and are presented in an enlarged view in Fig. 18.

Using the accuracy evaluation methodology, we calculated both the distribution of coordinate errors across specified intervals and the overall positional accuracy of the extracted shoreline. The computational efficiency of each algorithm was evaluated under a consistent hardware configuration (Windows 7, 64-bit; Intel[1] Core[2] i7-4790 CPU; 8 GB, RAM).

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[2]Trademarked.

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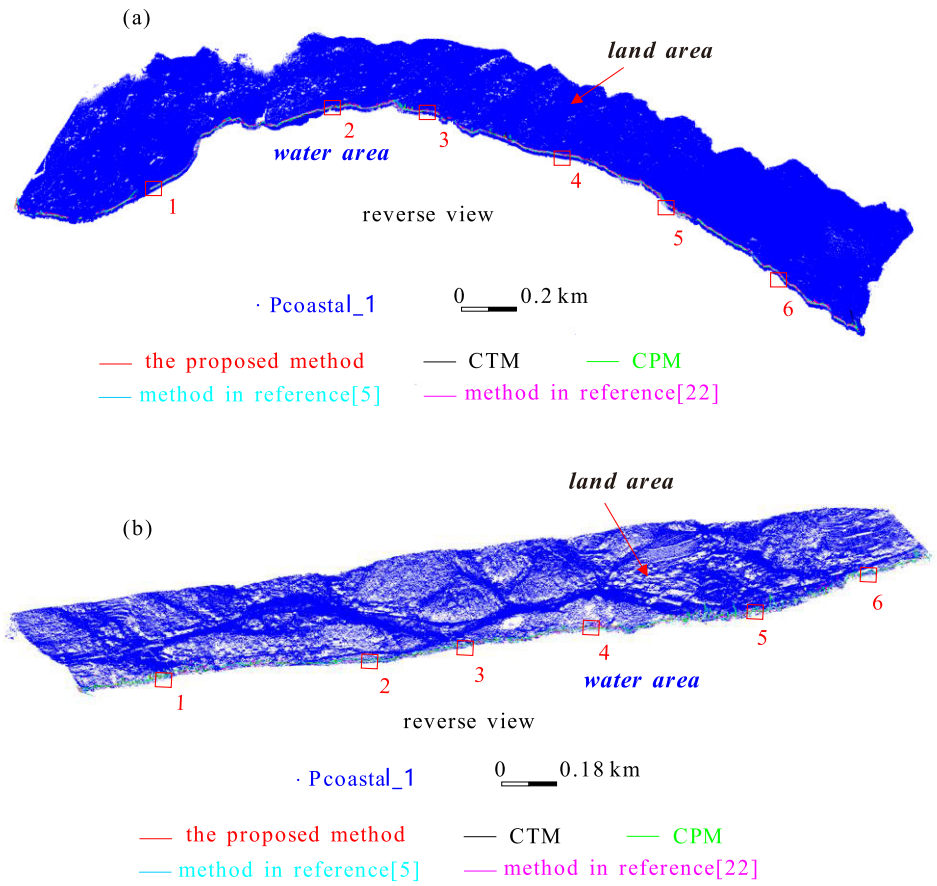


Fig. 19. Comparison of shoreline extraction results from the first five different methods overlaid on sea-area point clouds: (a) data 2 and (b) data 3.

TABLE II
STATISTICAL COMPARISON OF *Y*-COORDINATE ERRORS IN SHORELINE EXTRACTION ACROSS SEVEN METHODS

method	total no.of points	y-coordinate error interval (number of points)						standard deviation (m)	variance (m^2)	total run-time (s)
		>1	0.8 -1	0.6 -0.8	0.4 -0.6	0.2 -0.4	0- 0.2			
Proposed method	2459	**127**	**353**	531	562	612	274	**0.2438**	**0.0594**	**165**
CTM	2459	258	492	616	576	368	149	0.6632	0.4398	262
CPM	2459	316	627	729	**458**	**253**	**76**	0.7216	0.5207	248
ref. [5]	2459	216	462	**516**	548	486	231	0.5462	0.2983	228
ref. [22]	2459	228	478	572	506	458	217	0.6831	0.4666	258
U-Net	2459	209	458	583	516	524	169	0.5328	0.2839	216
PointNet ++	2459	231	492	549	489	516	182	0.5612	0.3149	206

The corresponding runtimes are quantitatively compared in Table II.

Explanation of chart annotations, red ellipse: indicates areas with significant jitter and instability. blue short line: denotes coarsely rendered regions lacking detailed representation. Only representative instances of jitter and coarse rendering are marked in each chart for clarity. As shown in Fig. 17(d), the green shoreline (CPM) significantly deviates from the other four shorelines, exhibiting poor consistency. In Fig. 17(f), the CPM shoreline traverses the area with excessive simplicity, failing to capture detailed coastal features. Similar oversimplifications are observed in Fig. 17(b) and (c). The

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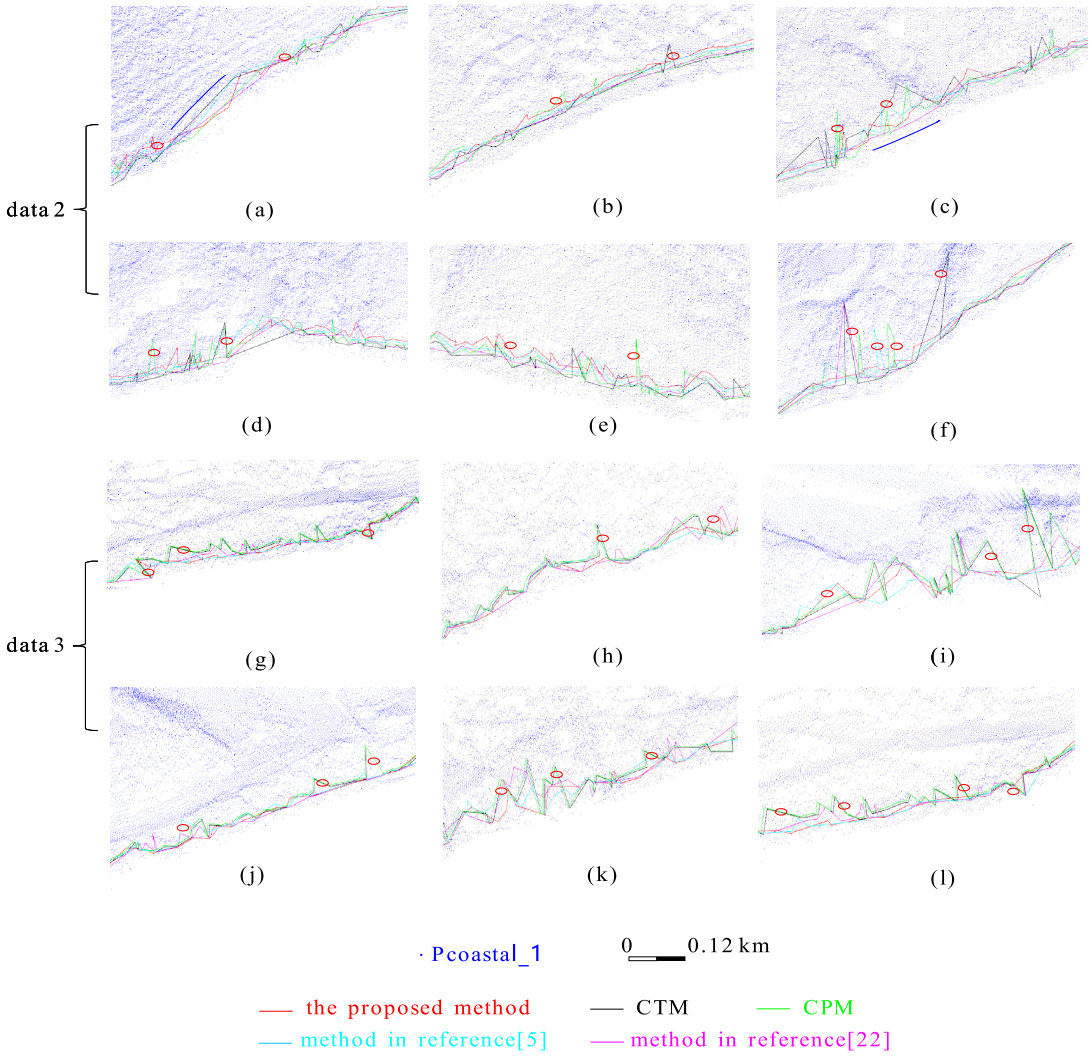


Fig. 20. High-resolution comparison of shoreline extraction methods: (a)–(f) detail views of marked regions 1–6 in Fig. 19(a) (Data 2); (g)–(l) detail views of marked regions 1–6 in Fig. 19(b) (Data 3).

cyan shoreline (see [5]) also demonstrates simplistic crossings in Fig. 17(b), (d), and (f), lacking the fine linear details present in other methods. Comparing the pink shoreline (see [22]) and the red shoreline (proposed method) in Fig. 17(a), (b), (d), and (f), the former exhibits more pronounced local shaking and abrupt jumps, both in frequency and magnitude. Fig. 17(c) and (e) shows some similarity between the pink and red shorelines, although the latter exhibits two abrupt jumps. The CTM (black shoreline) maintains reasonable directional and shape fidelity, though its positional deviation from the red and pink shorelines is notable. The annotations in Fig. 17 provide further comparative insights. In Fig. 17(c), the blue short line corresponds to the green shoreline derived from the CPM method, while the cyan shoreline (see [5]) is represented by a straight segment, both failing to capture the directional details of the shoreline in this area. As shown across the panels of Fig. 18, significant jitter and instability are observed in multiple sections of the blue shoreline (U-Net) and the pink shoreline (PointNet++). In addition, Fig. 17(d) reveals that extended portions of the blue shoreline lack refined morphological detail, further illustrating limitations in capturing continuous shoreline features. Table II further supports the superiority of the proposed method: it yields fewer high-error points and more low-error points compared to other methods. In terms of overall standard deviation and variance, the proposed method achieves the lowest values, indicating the highest accuracy. Specifically, compared to the other six methods with the lowest accuracy, it can be considered that this method has improved the overall accuracy of the shoreline by nearly 0.4778 m. In summary, the red shoreline (proposed method) demonstrates superior visual quality in Figs. 17 and 18, and higher accuracy in Table II compared to the green, cyan, pink, and black shorelines. According to the total run-time results in Table II, the proposed method achieves the shortest processing time. Computational demands are primarily concentrated in the coastal point cloud re-extraction stage, which involves processing the original large-scale coastal point cloud. Once this stage is complete, the data volume of the subsequent sea-area point cloud is substantially reduced, enabling highly efficient boundary extraction and processing. It should be noted that while the later stage of our method involves correction, adjustment, and fusion using

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TABLE III
Y-COORDINATE ERROR STATISTICS FOR SEVEN SHORELINE EXTRACTION METHODS (DATAS 2 AND 3)

data	method	total no.of points	y-coordinate error interval (number of points)						standard deviation (m)	variance (m^2)	total run-time (s)
			>1	0.8 -1	0.6 -0.8	0.4 -0.6	0.2 -0.4	0 -0.2			
data 2	proposed method	3158	**218**	**462**	**758**	778	631	311	**0.3162**	**0.0999**	**228**
	CTM	3158	328	585	856	828	**268**	293	0.7856	0.6171	368
	CPM	3158	353	556	928	726	283	312	0.7927	0.6283	326
	ref. [5]	3158	324	576	880	**658**	362	358	0.6256	0.3913	278
	ref. [22]	3158	312	528	896	785	331	306	0.6568	0.4313	263
	U-Net	3158	294	492	816	726	568	**256**	0.6156	0.3789	319
	PointNet ++	3158	378	576	796	689	585	334	0.5962	0.3553	282
data 3	proposed method	2166	**196**	**328**	454	482	468	238	**0.2853**	**0.0813**	**127**
	CTM	2166	216	454	672	428	**228**	168	0.5462	0.2983	185
	CPM	2166	246	553	526	**336**	378	127	0.5636	0.3176	178
	ref. [5]	2166	227	531	463	353	485	107	0.4878	0.2379	156
	ref. [22]	2166	217	481	**362**	568	362	176	0.4358	0.1899	162
	U-Net	2166	254	429	524	462	389	**103**	0.4954	0.2454	149
	PointNet ++	2166	231	458	468	379	428	202	0.4658	0.2169	153

the combined CPM, these operations are performed on the re-extracted sea-area point cloud. This eliminates the need for large-scale, exponentially complex iterative detection at this phase. As shown in Table II, the proposed method reduces the maximum run time by 97 s, demonstrating significant operational efficiency improvement. Therefore, the proposed shoreline extraction method outperforms the other six methods in both visual fidelity and precision, while also being more computationally efficient.

C. Model Generalization Experiment

To evaluate the model's effectiveness, accuracy, and generalization capability in extracting shorelines across different regions, this study conducted experiments using two airborne LiDAR point cloud datasets from Longkou City, Shandong Province. The first dataset (data2) has a size of 32.72 MB, containing 2 096 319 points, while the second dataset (Data 3) comprises 525 910 points (12.31 MB). The first five shoreline extraction methods were applied to both datasets, and the results were superimposed on the original point cloud to generate (see Fig. 18). For detailed analysis, six local regions in Fig. 19(a) and (b) were enlarged and presented in Fig. 20. The accuracy of each method was quantified using a coordinate error metric, with the results summarized in Table III.

Four representative areas were randomly selected from the comparative results of Data 2 and Data 3 for the proposed method, U-Net, and PointNet++, and are presented as enlarged views in Fig. 21.

As shown in Fig. 20(c)–(f), (h), (i), and (k), the black (CTM) and green (CPM) shorelines exhibit noticeable fluctuations and abrupt jumps at multiple locations, indicating a lack of smoothness and insufficient representation of fine-scale details. These discontinuities lead to significant positional errors in individual points, thereby reducing the overall accuracy of the shoreline delineation. Similarly, the pink shoreline (method from [22]) displays pronounced shaking and jumps in Fig. 20(h) and (k), while the cyan shoreline (method from [5]) also shows minor but observable fluctuations in Fig. 20(d) and (f). In contrast, the red shoreline (proposed method) remains consistently smooth and stable across all subfigures in Fig. 20, without abrupt transitions or irregularities. In addition, it captures finer curvatures and local variations more effectively, enhancing the representation of small-scale coastal features and reducing roughness or positional errors. Fig. 21 further illustrates that both advanced U-Net and PointNet++ exhibit noticeable jitter in their output shorelines across Data 2 and Data 3, compromising shoreline continuity and smoothness. In addition, as shown in Fig. 21(h), the blue shoreline generated by U-Net displays an elongated straight segment

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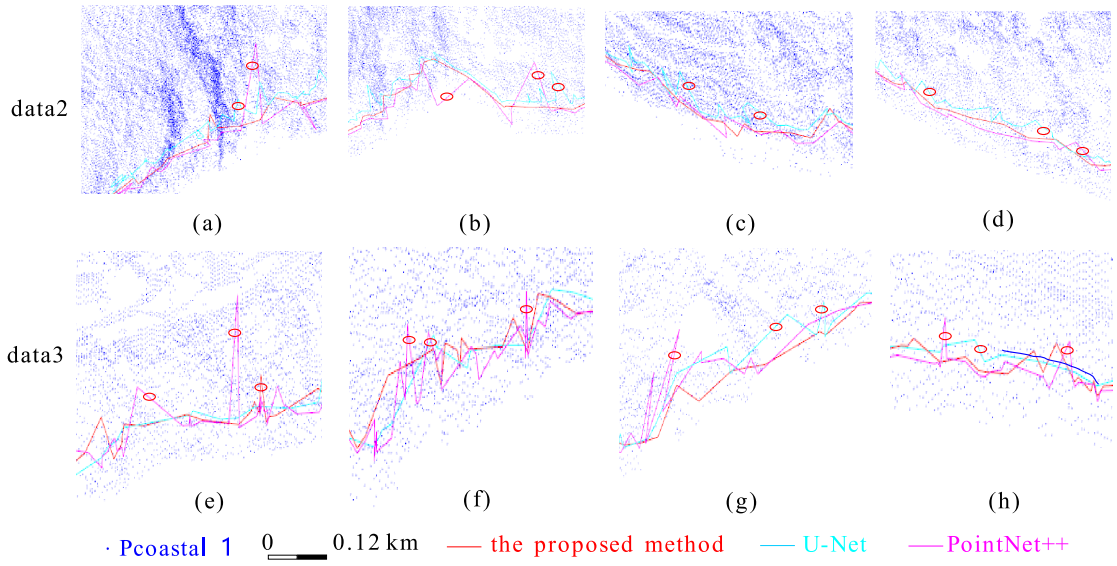


Fig. 21. Comparative results of the proposed method against U-Net and PointNet++. (a)–(d) Enlarged views of four randomly selected regions from Data 2. (e)–(h) Corresponding enlargements from four randomly selected regions in Data 3.

that fails to capture the natural undulations and wave-induced trends of the actual shoreline. The coordinate error data in Table III further support these observations. Compared to other methods, our approach yields fewer points in the high-error range and more points in the low-error range. Moreover, the standard deviation and variance of the shoreline positions are the smallest for our method in both tested datasets (Data 2 and Data 3). Specifically, our method improves the overall shoreline accuracy by 0.4765 m (Data 2) and 0.2773 m (Data 3), demonstrating its superior performance in reducing positional uncertainty. In terms of processing efficiency, it reduced processing time by 140 s (Data 2), and 58 s (Data 3), respectively, compared to the most time-consuming method. This comparative analysis demonstrates the proposed model's robustness, precision, and lightweight in diverse coastal environments.

IV. Conclusion

The Shoreline_All model proposed in this study presents a novel approach distinct from existing methods. Specifically, it classifies coastal point clouds into land and water subsets by applying a threshold derived from the laser echo intensity values of dominant land features in coastal zones. Each boundary point cloud is accurately extracted and individually fused. Subsequently, the fused point cloud coordinates are optimized based on coastal elevation characteristics. By integrating tidal data and coastal conversion models, the fused point cloud is transformed into a shoreline point cloud. Furthermore, the CPM is employed to detect and eliminate erroneous point clouds, which are then corrected, adjusted, and merged with the computed coastal point cloud to yield a more refined representation. To enhance completeness, a curve-fitting interpolation method is designed to fill gaps in the coastal point cloud. Experimental results demonstrate that the proposed method surpasses current mainstream techniques in both extraction performance and accuracy while exhibiting robust generalization capabilities. This approach is adaptable to diverse coastal types, offering a new technological solution for shoreline extraction. Its applications in ocean management, surveying, and related tasks hold significant practical value.

1) During Δd and Δh setup, efforts should focus on minimizing the re-extraction of coastal point clouds and reducing the inclusion of erroneous point clouds caused by isolated features on land. If such artifacts persist, manual removal may be necessary. The calibration accuracy of laser echo intensity values for different objects—particularly typical coastal features—requires further improvement. This includes refining data acquisition across multiple platforms and measurement modes. In addition, the relationship between echo intensity and water depth needs systematic calibration to enhance shoreline extraction precision.

2) While this method has demonstrated high accuracy, lightweight processing, and strong robustness in extracting results from the majority of vegetated coastal areas and all other coastal types, it still faces limitations in tall, dense mangroves. The dense canopy often blocks signals, leading to sparse or missing ground data. This lack of large-scale ground points hinders accurate subcanopy information retrieval, increasing shoreline extraction errors. To address this, terrestrial laser scanning (TLS) or shipborne LiDAR can be integrated to fill data gaps, which can then be registered and fused with airborne point clouds for complete area coverage.

3) While airborne LiDAR point clouds offer a promising approach for shoreline extraction, their development history remains shorter compared to traditional aerial imagery techniques. Meanwhile, interferometric synthetic aperture radar (InSAR) has advanced rapidly. Future research should explore integrating airborne LiDAR, aerial imagery, and InSAR to achieve higher accuracy shoreline extraction.

Data Availability Statement

The data that support the findings of this study are available upon reasonable request from the authors.

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